Description

Star tracker

This project is an automated camera mount designed for long-exposure astrophotography. By continuously rotating opposite to the Earth's rotation, the device tracks a specific part of the sky to keep celestial objects perfectly framed. Without tracking, a camera using a standard 50mm lens is limited to 10-second exposures before the Earth's movement causes visible star trails.

The system targets objects using input celestial coordinates or by being manually aimed at a desired location. To calculate the tracking movements, the device pulls its exact latitude and real-time UTC from a GPS module, and aligns itself using a magnetometer to find North. System converts Equatorial (RA, Dec) Coordinates to Horizontal (Altitude, Azimuth) Coordinates. The physical movement is driven by two 180-degree servos. The coordinates could be given by entering data , or manually rotating of servos with potentiometers.

Requirements

Initialization

IR-1 System shall find its location and UTC

IR-2 System shall find North direction

Tracking sky requirements

TR-1 System shall receive equatorial coordinates from user.

TR-2 System shall follow chosen part of the sky.

Safety requirements

SR-1 IF provided coordinates are below horizon, THEN system shall not rotate there